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Subject Name : Machine Learning

Unit No: 1

Unit Name : Introduction to Machine Learning & Data Preprocessing

Faculty Name : Dr. Ritu Jain

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Unit No.: 1.2

Faculty Name :

Batch and Online Learning

- Another criterion used to classify Machine Learning systems is
  - **whether or not the system can learn incrementally from a stream of incoming data.**

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Batch Learning (Offline Learning)

- Batch Learning, also known as **offline learning**, is a traditional machine learning method where the model is trained on a complete, static dataset all at once.
- Once trained, the model is deployed for use and does not continue to learn or update from new data in real-time.
  - **Cannot learn incrementally** — no learning after deployment
  - Once deployed, the system **only makes predictions**, it does not update itself

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Batch (Offline) Machine Learning Workflow

1. **Data Collection & Ingestion:** complete static dataset is gathered for analysis.
2. **Preprocessing & Feature Engineering:** The entire dataset is preprocessed and feature engineered
3. **Data Splitting:** The processed dataset is divided into training, validation, and test sets.
4. **Model Training**: Algorithms are trained from scratch on the full training set.
5. **Model Evaluation & Validation**: Performance is assessed on validation/test sets using metrics, tuned if needed.
6. **Model Deployment & Periodic Retraining:** The finalized model is deployed for inference; it is periodically retrained on new batches to adapt to data shifts, with monitoring for degradation.

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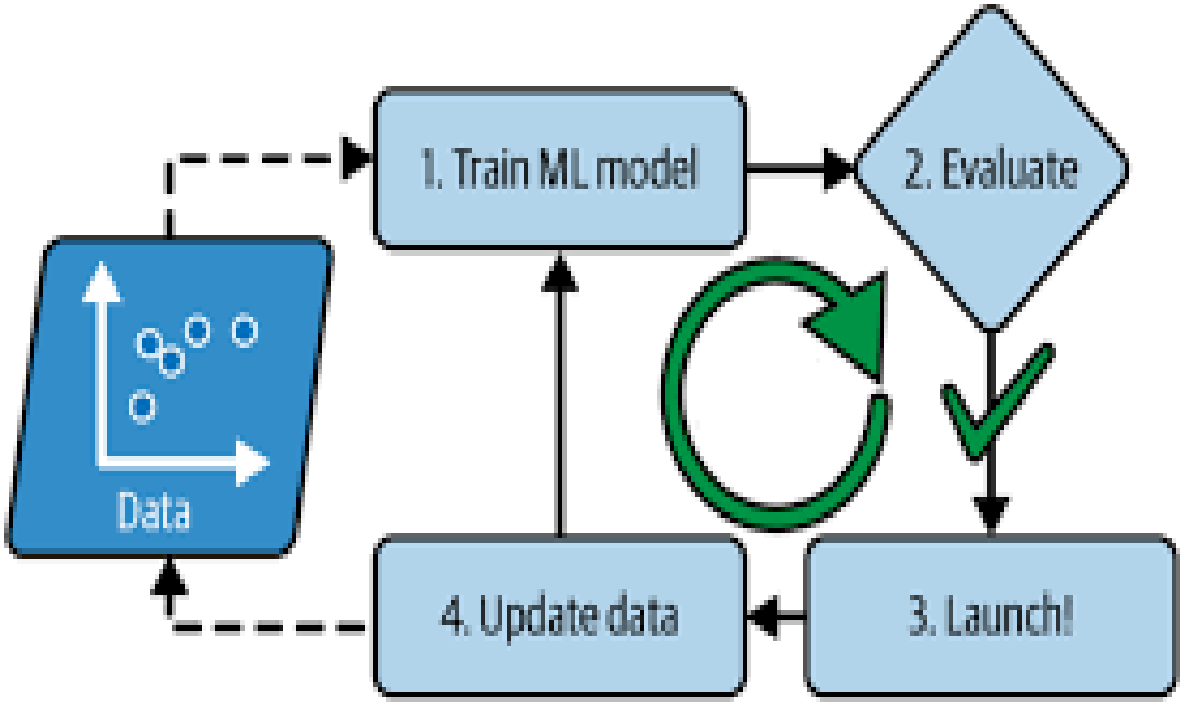
Batch Learning



Key Characteristics of Batch Learning

- **Training Process:** The entire available dataset is fed into the learning algorithm at once in an offline environment.
  - The model's parameters are updated during this single, extensive training phase.
- **Model Updates:** The model remains static after deployment.
  - To incorporate new data, the entire model must be retrained from scratch using both the old and new data, and then the updated model replaces the previous one in production.
- **Resource Consumption:** Training with large batches of accumulated data is computationally intensive, requiring significant CPU, memory, and disk I/O resources and time.

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Key Characteristics of Batch Learning

- **Adaptability**: It has *limited* adaptability to new incoming data or sudden changes in data patterns.
  - The model's performance may degrade over time as the real-world environment evolves.

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Advantages of Batch Learning

- **Stability**: Once trained, the model is stable and predictable, as it is not impacted by individual noisy or non-representative data points in production.
- **Accuracy**: Training on the entire dataset allows the model to capture global patterns and can result in high accuracy for stable problems.
- **Simplicity**: The process is generally easier to implement, manage, and validate, as engineers can thoroughly test the model before deployment.
- **Reproducibility**: The static nature of the training data and process makes the results easier to reproduce and audit.

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Disadvantages of Batch Learning

- **Slow Adaptation**: It cannot adapt to new data in real-time, leading to delays in responding to new trends or conditions.
- **Resource-Intensive Retraining**: Retraining the entire model from scratch every time new data arrives is computationally expensive and time-consuming.
  - Frequent retraining can become **very expensive**
- **Downtime**: The deployment process for an updated model might require temporary downtime.
  - Full retraining can take **hours or days**
  - Not suitable for **rapidly changing environments** (e.g., stock market prediction)

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When to use Batch learning

- Batch learning is the industry standard for scenarios where data does not change rapidly.
  - **Stable Environments:** Use for tasks like Image Recognition (e.g., identifying animals) where the underlying patterns remain consistent for years.
  - **High Precision Tasks:** Ideal for high precision task where model stability and thorough validation are critical before deployment.
  - **Resource Constraints in Production:** Best when the production environment (like a remote sensor or a mobile app) lacks the compute power to handle real-time training.

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Better Alternative When Data Changes Frequently

- Use **Online / Incremental Learning**
  - Model learns **continuously from new data**
  - Suitable for **dynamic, real-time systems**

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Online Learning (Incremental Learning)

- *method where models are updated incrementally, one data point or small batch at a time, allowing for real-time adaptation to new, streaming data.*
- Online learning is computationally efficient, resource-friendly, and ideal for scenarios where data evolves over time, such as in recommendation systems, fraud detection, and financial price prediction.

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Online Machine Learning Workflow

1. **Data Ingestion & Streaming:** Data is collected in real-time from sources like IoT devices, web user interactions, or market feeds.
2. **Preprocessing & Feature Engineering:** Incoming data is immediately cleaned, transformed, and processed into suitable features on-the-fly.
3. **Incremental Model Updating:** Instead of retraining from scratch, the current model updates its parameters using only the new data points (online learning algorithms like Stochastic Gradient Descent).
4. **Prediction & Inference**: The updated model instantly makes predictions on new data.
5. **Performance Evaluation & Monitoring:** The model's accuracy is continuously monitored against incoming actual results (feedback), with automated alerts for performance drift.
6. **Model Adaptation:** The system adapts to changes in data distribution by continually learning, ensuring the model remains accurate over time.

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Online Machine Learning

- **Key Parameter: Learning Rate**
  - Controls how fast the system adapts to new data:
    - **High learning rate**
      - Learns quickly
      - Adapts fast to changes
      - May **forget old patterns too quickly**
    - **Low learning rate**
      - Learns slowly
      - Remembers old data longer
      - Less sensitive to **noise or unusual data**

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- When the dataset is **too large to fit into memory**, instead of training once on the entire dataset (batch learning), we:
  - **Split the data into small chunks**
  - **Train the model incrementally**, one chunk at a time
  - Continuously **evaluate and improve** the model.
- **2. Chop into pieces**
- The big dataset is **split into small chunks** (also called:
  - mini-batches
  - data streams
  - data instances)
- Each chunk is small enough to be processed quickly.

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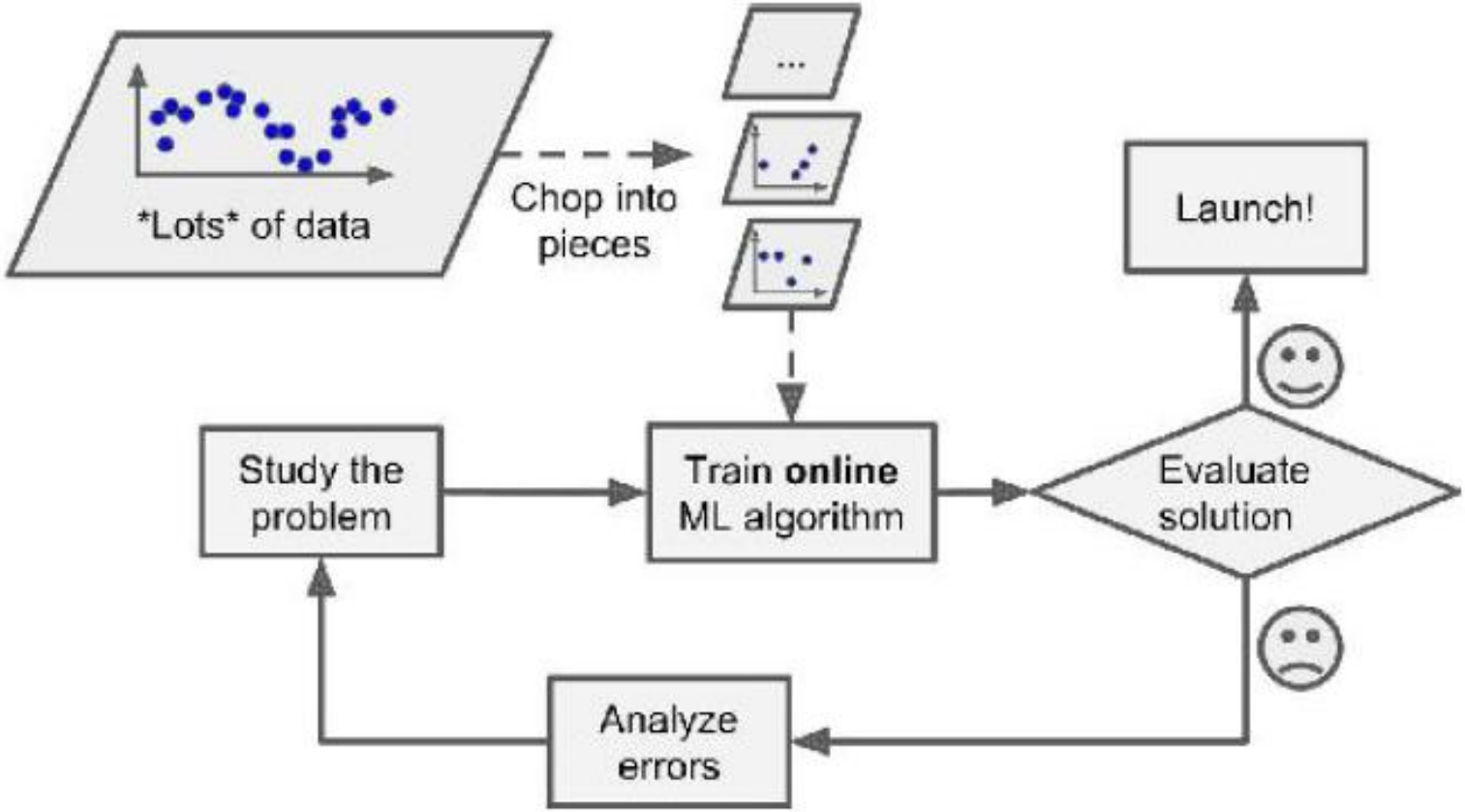

Figure 1-14. Using online learning to handle huge datasets

Benefits

- **Real-time adaptation:** Models adjust instantly to new patterns.
- **Efficiency**: Eliminates the need for costly, large-scale retraining on historical data.
- **Scalability**: Suitable for high-velocity data streams.
- **Efficient Use of Resources:** After learning from data, the system can **discard it**
  - Saves **large amounts of storage space**
  - Good choice when **computing resources are limited**
- Handling Very Large Datasets (**Out-of-Core Learning**):
- Used when the dataset **does not fit into main memory**
  - Process:
    - Load a portion of data
    - Train the model on that portion
    - Repeat until all data is processed
- Makes learning possible on **massive datasets**

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Where Online Learning is Useful

- Systems with **continuous data streams**
  - Stock prices
  - Sensor data
  - User behavior logs
- Situations requiring **rapid adaptation**
- Systems that must learn **autonomously in real time**

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Challenges in Online Learning

- System can be harmed by **bad or noisy data**
- Performance may **gradually degrade**
- Risk sources:
  - Faulty sensors
  - Malicious or spam inputs
  - Abnormal data patterns

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How to Reduce Risk

- Continuously **monitor system performance**
- Detect performance drops early
- Temporarily **stop learning** if needed
- Revert to a **previous stable model**
- Use **anomaly detection** to filter suspicious data

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Summary

- Online learning is ideal for:
  - Real-time adaptation
  - Streaming or massive data
  - Resource-constrained environments
- But it requires **careful monitoring and tuning** to stay reliable.

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Batch Learning vs Online Learning

Feature	**Batch Learning**	**Online Learning**
Training Style	Trained on **entire dataset at once**	Trained **incrementally** on new data
Learning After Deployment	No learning after release	Continues learning
Model Updates	Requires **full retraining from scratch**	Updates happen **step-by-step**
Speed of Adaptation	Slow to adapt	Fast adaptation to new trends
Computational Cost	High (CPU, memory, storage)	Lower per update
Data Storage Need	Must store **all historical data**	Can discard old data after learning
Best For	Stable environments with **slow data change**	Dynamic environments with **rapid data change**
Handles Huge Datasets	Difficult if data is too large	Supports **out-of-core learning**
Risk Factors	Expensive retraining	Sensitive to **noisy or malicious data**
Monitoring Need	Less frequent monitoring	Requires **continuous monitoring**

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When to Choose What?

- **Choose Batch Learning when:**
  - Data changes slowly
  - High computational resources are available
  - Model updates are infrequent
- **Choose Online Learning when:**
  - Data arrives continuously
  - Real-time or fast adaptation is required
  - Storage or memory is limited

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Model-Based Learning

- **What is Model-Based Learning?**
  - **Model-Based Learning** focuses on building a mathematical or algorithmic model that generalizes the patterns and relationships within the training data. This model is then used to make predictions on new, unseen data.
- **How It Works:**
  - A model is trained using the dataset.
  - The model learns the underlying patterns and relationships.
  - Predictions are made based on this learned knowledge, without relying on the original training data.

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Model-Based Learning

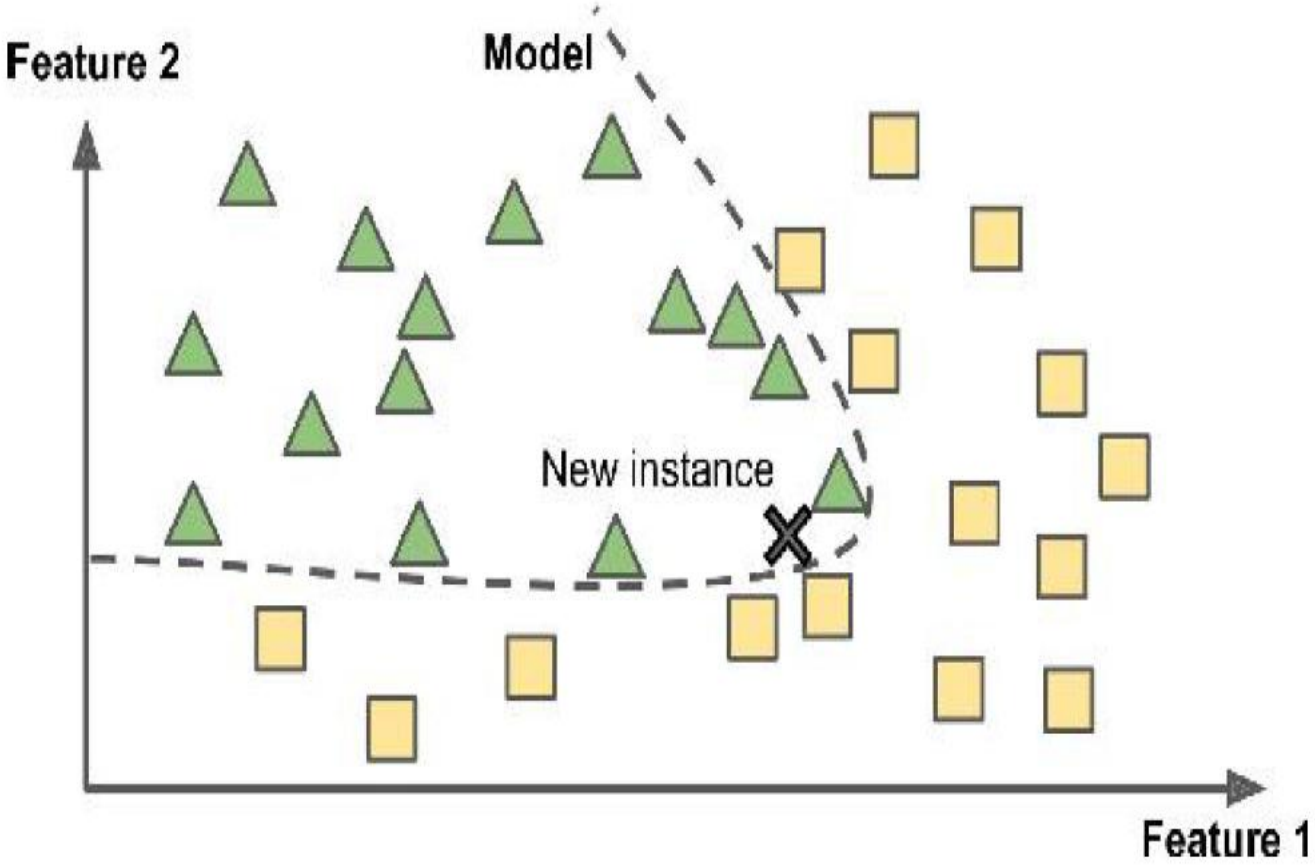

Figure 1-16. Model-based learning

Model-Based Learning

- **Examples:**
  - **Linear Regression:** Establishes a linear relationship between input and output.
  - **Decision Trees:** Creates a tree structure to make decisions based on feature splits.
  - **Neural Networks:** Mimics the human brain to identify patterns in complex data.
- **Advantages:**
  - Faster predictions after training.
  - Memory-efficient, as it doesn't require storing the entire dataset.
  - Suitable for large datasets and complex problems.
- **Limitations:**
  - Requires retraining for new data.
  - Model accuracy depends on the quality and representativeness of the training data.

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Instance-Based Learning

- **What is Instance-Based Learning?**
  - Instance-Based Learning, on the other hand, memorizes the training data and uses it directly for making predictions.
  - It does not generalize or create a standalone model but relies on comparing new data points to stored instances.

- **How It Works:**
  - Training data is stored as-is.
  - Predictions are made by comparing new data points to the closest or most similar instances using distance metrics like Euclidean distance.

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Instance-Based Learning

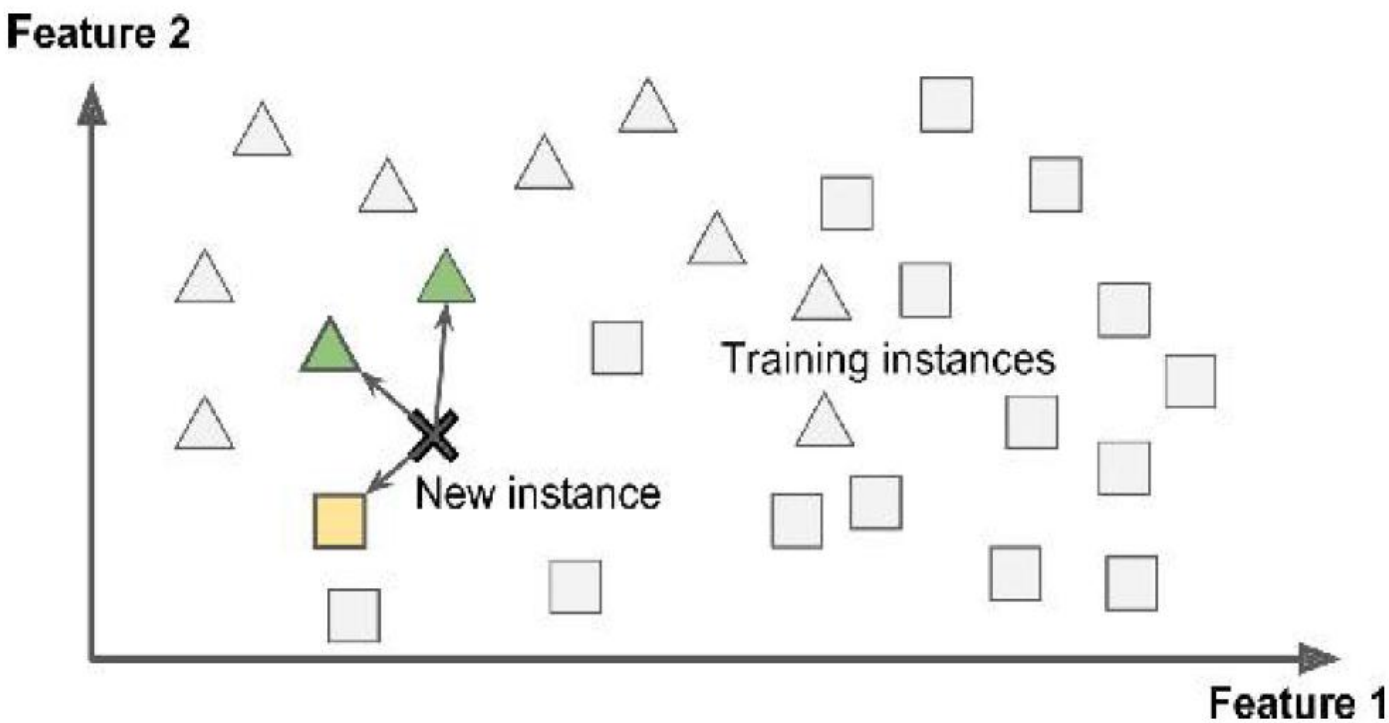

Figure 1-15. Instance-based learning

Instance-Based Learning

- **Examples:**
  - **k-Nearest Neighbors (k-NN):** Identifies the 'k' closest instances to predict the outcome.
  - **Support Vector Machines (SVM):** Uses specific data points (support vectors) to define boundaries.
- **Advantages:**
  - Easy to implement and understand.
  - Adapts naturally to new data since no retraining is needed.
- **Limitations:**
  - Computationally expensive for large datasets due to similarity calculations.
  - Storage-intensive as it requires keeping all training data.

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Machine Learning Life Cycle

- Machine Learning Lifecycle is a structured process that defines how machine learning (ML) models are developed, deployed and maintained.
- It consists of a series of steps that ensure the model is accurate, reliable and scalable.
- It includes defining the problem, collecting and preparing data, exploring patterns, engineering features, training and evaluating models, deploying them into production and continuously monitoring performance to handle issues like data drift and retraining needs.

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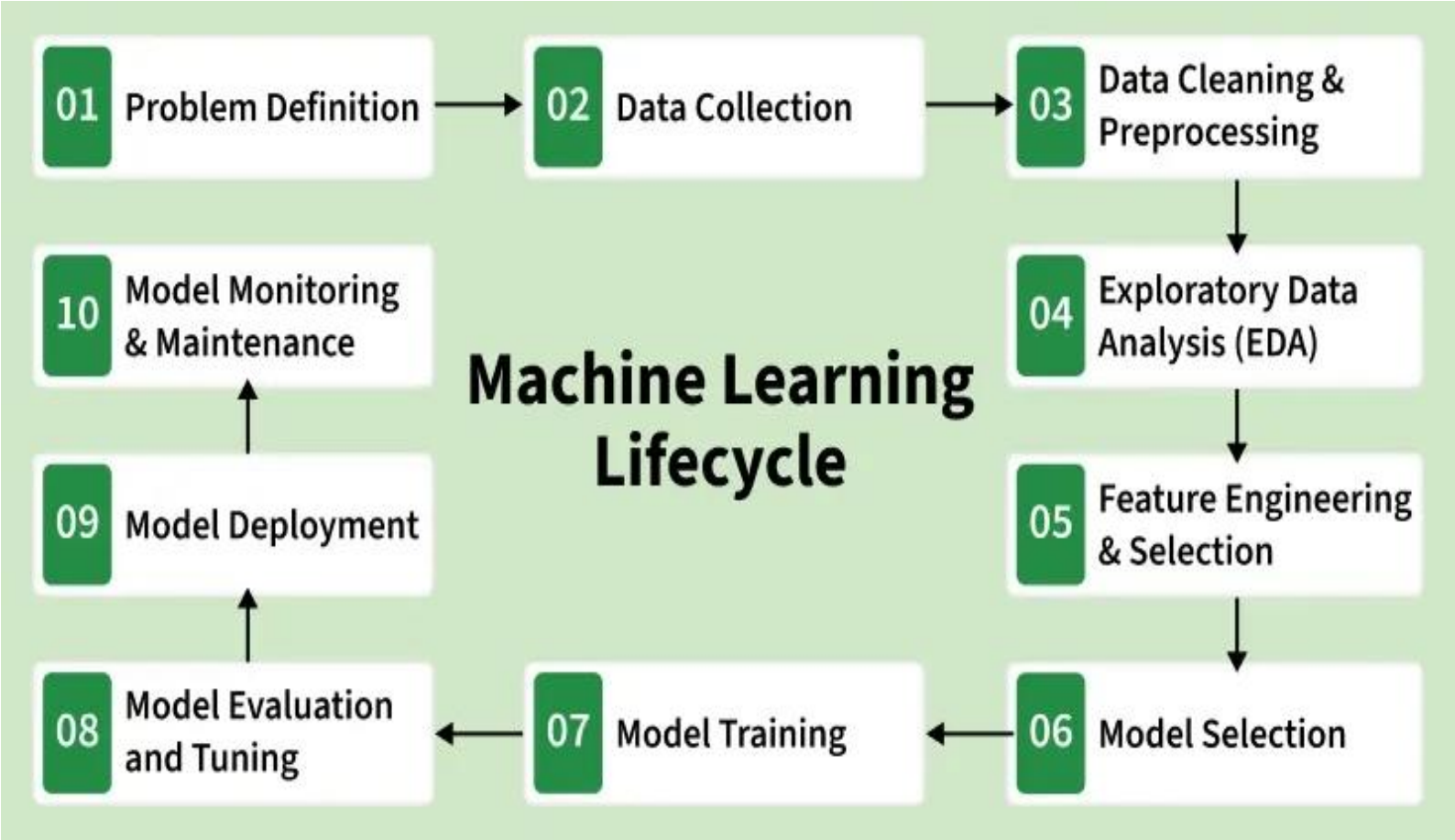
Machine Learning Life Cycle



Machine Learning Life Cycle

Step 1: Problem Definition

- The first step is identifying and clearly defining the business problem.
- A well-framed problem provides the foundation for the entire lifecycle.
- Important things like project objectives, desired outcomes and the scope of the task are carefully designed during this stage.
  - Collaborate with stakeholders to understand business goals
  - Define project objectives, scope and success criteria
  - Ensure clarity in desired outcomes
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Machine Learning Life Cycle

Step 2: Data Collection

- Once your problem is defined, the next step is to collect the data that will feed your model.
- This phase focuses on gathering relevant, high-quality, and diverse data from the right sources.
- The type, quantity, and variety of your data directly impact the model's ability to learn and perform well.

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Machine Learning Life Cycle

Step 3: Data Cleaning and Preprocessing

- Raw data is often messy and unstructured and if we use this data directly to train then it can lead to poor accuracy.

Step 4: Exploratory Data Analysis (EDA)

- Exploratory Data Analysis (EDA) is used to uncover insights and understand the dataset's structure.
- During EDA patterns, trends and insights can be observed which may not be visible by naked eyes.
- This valuable insight can be used to make informed decision.

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Machine Learning Life Cycle

Step 5: Feature Engineering and Selection

- Feature engineering and selection is a process that involve selecting only relevant features to enhance model efficiency and prediction while reducing complexity.

Step 6: Model Selection

- For a good machine learning model, model selection is a very important part as we need to find model that aligns with our defined problem, nature of the data, complexity of problem and the desired outcomes.

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Machine Learning Life Cycle

Step 7: Model Training

- With the selected model the machine learning lifecycle moves to model training process. This process involves exposing model to historical data allowing it to learn patterns, relationships and dependencies within the dataset.

Step 8: Model Evaluation and Tuning

- Model evaluation involves rigorous testing against validation or test datasets to test accuracy of model on new unseen data. It provides insights into model's strengths and weaknesses. If the model fails to acheive desired performance levels we may need to tune model again and adjust its hyperparameters to enhance predictive accuracy.

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Machine Learning Life Cycle

Step 9: Model Deployment

- Now model is ready for deployment for real-world application. It involves integrating the predictive model with existing systems allowing business to use this for informed decision-making.

Step 10: Model Monitoring and Maintenance

- After Deployment models must be monitored to ensure they perform well over time. Regular tracking helps detect data drift, accuracy drops or changing patterns and retraining may be needed to keep the model reliable in real-world use.

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